

OSI 7-Layer Model Assignment

Comprehensive Solutions and Detailed Analysis

Task 1: OSI 7-Layer Model Diagram & Layer Breakdown

The Open Systems Interconnection (OSI) model is a conceptual framework that standardizes the functions of a telecommunication or computing system without regard to its underlying internal structure and technology. It divides network communication into seven distinct abstraction layers, from the physical hardware up to the user application.

Layer #	Layer Name	Primary Function	Real-World Example
7	Application	Direct interface for user applications to network services.	HTTP, HTTPS, DNS, FTP, SMTP
6	Presentation	Data translation, encryption, decryption, and compression.	SSL/TLS, JPEG, ASCII, MPEG
5	Session	Establishes, maintains, and terminates communication sessions.	NetBIOS, RPC, PPTP
4	Transport	End-to-end reliable/unreliable data delivery, flow control.	TCP, UDP
3	Network	Logical addressing, packet forwarding, and routing.	IP (IPv4/IPv6), Routers, ICMP
2	Data Link	Node-to-node transfer, framing, physical addressing (MAC).	Switches, Ethernet (IEEE 802.3), MAC Addresses
1	Physical	Transmission and reception of raw bit streams over physical medium.	Hubs, Network Cables (Cat6), Fiber Optics

Task 2: Networking Scenarios Matching

Matching each networking scenario to its correct OSI layer, name, and layer number following the requested format ('Scenario: Layer Name (Layer Number)'):

- **a) Encrypting data before sending:** Presentation Layer (Layer 6)
- **b) Routing packets across networks:** Network Layer (Layer 3)
- **c) Converting data to electrical signals:** Physical Layer (Layer 1)
- **d) Opening a web page in your browser:** Application Layer (Layer 7)
- **e) Detecting and correcting errors in frames:** Data Link Layer (Layer 2)

Task 3: Daily Application Analysis (WhatsApp)

To understand how the OSI model functions in real-world software, consider WhatsApp—a globally used messaging and VoIP application. When a user sends a multimedia message or text to a contact, multiple OSI layers collaborate seamlessly:

- 1. Application Layer (Layer 7):** Layer 7 provides the direct user interface and executes protocols like HTTPS and XMPP/MQTT variations to initiate message composition, handle contact lookups, and format application-level payloads.
- 2. Presentation Layer (Layer 6):** Layer 6 is responsible for end-to-end encryption (WhatsApp uses the Signal Protocol). Before text or media leaves the device, this layer encrypts the plaintext payload into ciphered text and standardizes media formats (e.g., JPEG for images, Opus for voice notes) so the recipient's device can correctly parse them.
- 3. Transport Layer (Layer 4):** Layer 4 ensures reliable, ordered, and error-checked delivery of message segments using TCP (Transmission Control Protocol). It manages port numbers, tracks packet delivery acknowledgments, and resends lost packets if network drops occur during transmission.

Task 4: Troubleshooting Scenario Analysis

Given Scenario: You are unable to access a website via a web browser (e.g., getting a DNS resolution error, HTTP 404/500 status, or blank page), but you can successfully ping its IP address.

Identified Problem Layers & Reasoning:

Diagnostic Baseline: Since you can successfully ping the destination IP address, it proves that Layers 1 (Physical), 2 (Data Link), and 3 (Network) are fully functional. Packets are successfully traversing the physical media, local framing is correct, and routers are successfully forwarding ICMP echo requests/replies across networks. Therefore, the failure occurs at higher layers:

- **Application Layer (Layer 7):** If you pinged an IP address directly (or if domain name resolution succeeded via ping), but the browser fails to load the web page, the issue may lie at Layer 7. Common causes include an incorrect URL path, an unhandled web service status error

(e.g., 502 Bad Gateway or 403 Forbidden), or the web server daemon (like Nginx or Apache) being down while the underlying server machine remains powered on and responding to ping.

- **Transport Layer (Layer 4):** A successful ping verifies network connectivity on ICMP, but the website requires TCP communication on specific ports (typically Port 80 for HTTP or Port 443 for HTTPS). If a stateful firewall, security group, or local software firewall is blocking TCP Port 443/80, or if the web service port is closed on the server side, the browser handshake will fail.

- **Presentation Layer (Layer 6):** If the connection requires secure encrypted sessions (HTTPS/TLS handshake), a failure in negotiating cryptographic certificates, cipher suites, or TLS version mismatch between client and server will terminate the connection before web data can be displayed, even though the raw IP connection is intact.